

Algebra (7 th std Term 3)

Choose the appropriate answer

1. If $a + b = 5$ and $a^2 + b^2 = 13$, then $ab = ?$
(a) 12 (b) 6 (c) 5 (d) 13
Ans: (b) 6
2. $(5 + 20) (-20 - 5) = ?$
(a) -425 (b) 375 (c) -625 (d) 0
Ans: (c) -625
3. The factors of $x^2 - 6x + 9$ are
(a) $(x - 3)(x - 3)$ (b) $(x - 3)(x + 3)$ (c) $(x + 3)(x + 3)$ (d) $(x - 6)(x + 9)$
Ans: (a) $(x - 3)(x - 3)$
4. The common factors of the algebraic expressions ax^2y , bxy^3 and $cxyz$ is
(a) x^2y (b) xy^2 (c) xyz (d) xy
Ans: (d) xy
5. The solutions of the inequation $3 \leq p \leq 6$ are (where p is a natural number)
(a) 4, 5 and 6 (b) 3, 4 and 5 (c) 4 and 5 (d) 3, 4, 5 and 6
Ans: (d) 3, 4, 5 and 6
6. The solution of the inequation $5x + 5 \leq 15$ are (where x is a natural number)
(a) 1 and 2 (b) 0, 1 and 2 (c) 2, 1, 0, -1, -2... (d) 1, 2, 3...
Ans: (a) 1 and 2
7. The cost of one pen is ₹ 8 and it is available in a sealed pack of 10 pens. If Shweta has only ₹ 500, how many packs of pens can she buy at the maximum?
(a) 10 (b) 5 (c) 6 (d) 8
Ans: (c) 6
8. The inequation that is represented on the number line as shown below is _____.

(a) $-4 < x < 0$ (b) $-4 \leq x \leq 0$ (c) $-4 < x \leq 0$ (d) $-4 \leq x < 0$
Ans: (d) $-4 \leq x < 0$
9. A symbol in algebra having a fixed value is called.
(a) Constant (b) Algebraic expression (c) Variable (d) None of these
Ans: (a) Constant
10. A symbol which can be given or assigned a varied number of numerical values is called
(a) Term (b) Variable (c) Constant (d) Coefficients
Ans: (b) Variable

11. A combination of constants and variables connected by the basic mathematical operators (+, -, ×, ÷) is called

- (a) Algebraic expression (b) Like terms (c) Unlike terms (d) Term

Ans: (a) Algebraic expression

12. In a polynomial the exponents of the variables are always

- (a) Integers (b) Positive integers (c) Non- positive integers (d) None of these

Ans: (b) Positive integers

13. Which of the following is a binomial?

- (a) $8 \times x \div x$ (b) $12a^2 + 7b + 5c$ (c) $5a \times 7b \times 8c$ (d) $12(a^3 + a)$

Ans: (d) $12(a^3 + a)$

14. The various parts of an algebraic expression connected by + or – sign are called

- (a) Expression (b) Constants (c) Term (d) None of these

Ans: (c) Term

15. Multiply $3x$ by $(2x + 5y)$

- (a) $6x^2 + 15xy$ (b) $6x^3 + 15xy^2$ (c) $6x + 15x^2y^2$ (d) None of these

Ans: (a) $6x^2 + 15xy$

16. Subtract $6a - 4b$ from $5a + 8b$ we get

- (a) $-a + 12b$ (b) $-a - 12b$ (c) $a + 12b$ (d) None of these

Ans: (a) $-a + 12b$

17. The sum of $a + b + ab$, $-b + c - bc$ and $-c - a + ac$ is

- (a) $2c + ab - bc + ac$ (b) $ab - bc - ac$
(c) $ab - bc + ac$ (d) $2a + 2b - 2c + ab - ac - bc$

Ans: (b) $ab - bc - ac$

18. Rule $(+x) \times (-y)$ becomes.

- (a) $+xy$ (b) $-xy$ (c) xy (d) None of these

Ans: (b) $-xy$

19. Terms having same literal factors are called

- (a) Coefficient (b) Like terms (c) Constant (d) None of these

Ans: (b) Like terms

20. $5a - 3[3a - (4 - 7a)] + 4(a - 3)$ on simplification is equal to

- (a) $-21a$ (b) $21a + 12$ (c) $-21a - 24$ (d) -24

Ans: (d) -24

21. $(3x - 4)(2x^2 - 5x + 1) - (2x - 1)(3x^2 + 7x - 5)$ equals.

- (a) $12x^3 - 40x^2 + 34x - 1$ (b) $34x^2 - 40x - 9$
(c) $-34x^2 + 40x - 9$ (d) $12x^3 - 34x^2 + 40x + 9$

Ans: (a) $12x^3 - 40x^2 + 34x - 1$

22. Divide $-54 \times 4y^3z$ by $6x^2y^2z$

- (a) $9xy^2$ (b) $-9x^2y$ (c) $9x^2y^2$ (d) $-9x^2y^2$

Ans: (b) $-9x^2y$

23. Simplify $(a+1)(a+2)(a+3)$ then

- (a) $a^3 + 6a^2 + 11a + 6$ (b) $a^2 + 6a^3 - 11a + 8$
(c) $a^4 + 6a^2 - 11a - 6$ (d) None of these

Ans: (a) $a^3 + 6a^2 + 11a + 6$

24. Which polynomial has the highest degree?

- (a) $5x^2 - 2x^8 + x^6$ (b) $-7x^2 + 20$ (c) $27x^{10} + 3x^5 - 16$ (d) $-150x^3$

Ans: (c) $27x^{10} + 3x^5 - 16$

25. By how much is $a^4 - 6a^2b^2 + b^4$ more than $a^4 + 4a^2b^2 + b^4$?

- (a) $-2a^2b^2$ (b) $2a^4 + b^4$ (c) $-10a^2b^2$ (d) $2a^2b^2$

Ans: (c) $-10a^2b^2$

26. What will be the missing term? $(-14y^2 + 9y^2 - 12y + 3) + (2y^2 + -6y - 2) = (-3y^2 - 15y + 1)$

- (a) $-6y$ (b) $3y$ (c) $-3y$ (d) $6y$

Ans: (b) $3y$

27. $(-7x^7y^3 - 56xy^5z^2) \div (x^7 - 8xy^2z^2)$

- (a) $-15y^3$ (b) $15y^3$ (c) y^3 (d) None

Ans: (d) None

28. The sum or difference of monomials is called

- (a) Polynomial (b) Term (c) Coefficient (d) None

Ans: (a) Polynomial

29. The term having different variable parts are called

- (a) Constant and variables (b) Terms (c) Like terms (d) Unlike terms

Ans: (d) Unlike terms

30. The sum of the exponents of the variables is:

- (a) Degree (b) Exponents (c) Variables (d) None

Ans: (a) Degree

31. $x^0 + y^0 + 3$ equals

- (a) 1 (b) 2 (c) 3 (d) 5

Ans: (d) 5

32. Which is the correct one

- (a) Dividend = Remainder $\times$ Q + Divisor (b) Divisor = Dividend $\times$ Q + R
(c) Dividend = Q $\times$ D + R (d) None of these

Ans: (c) Dividend = Q $\times$ D + R

33. When $2x^2 - 11x + 12$ divided by $x - 4$ then remainder is

- (a) 1 (b) 2 (c) 0 (d) 4

Ans: (c) 0

34. If a is greater than b which is correct

- (a) $a < b$ (b) $a > b$ (c) $a = b$ (d) $a \neq b$

Ans: (b) $a > b$

35. A number satisfies an inequality, if it makes that inequality
(a) False (b) Incorrect (c) True (d) None of these

Ans: (c) True

36. On multiplying both sides by the same negative number it becomes.
(a) Changed (b) Reversed (c) Un changes (d) None of these

Ans: (b) Reversed

37. The set of elements from which the replacement of the _____ is taken is called replacement Set.

(a) Variables (b) Constants (c) Operators (d) Numbers

Ans: (a) Variables

38. We should not _____ both sides of an inequality by a negative number unless it is desired to reverse the inequality.

(a) Multiply (b) Divide (c) Multiply or Divide (d) None of these

Ans: (c) Multiply or Divide

39. The set of all possible values of 'x' which satisfy a given in equation is called
(a) Solution set (b) Replacement set (c) Reverse set (d) None of these

Ans: (a) Solution set

40. If $3x + 2 \geq 14$ then x is

(a) $x < 4$ (b) $x > 4$ (c) $x \geq 4$ (d) None of these

Ans: (c) $x \geq 4$

41. On changing sides the _____ of the inequality is

(a) Reversed (b) Changed (c) Unchanged (d) None of these

Ans: (a) Reversed

42. Solve $3x + 7 < 10$ the x is

(a) $x < 1$ (b) $x \leq 1$ (c) $x \geq 2$ (d) None of these

Ans: (a) $x < 1$

43. Solve $x + 3 > 8$ where $x \in \mathbb{W}$ then

(a) $x = 1, 2, 3, \dots$ (b) $x = 6, 7, 8, \dots$ (c) $x = 0$ (d) None of these

Ans: (b) $x = 6, 7, 8, \dots$

44. What inequality is shown on the graph below?



(a) $2x > 6$ (b) $2x > -6$ (c) $2x \geq 6$ (d) $2x \geq -6$

Ans: (b) $2x > -6$

45. Which is the solution to $4x - 6 < 10$?

(a) $x > 1$ (b) $x < 1$ (c) $x > 4$ (d) $x < 4$

Ans: (c) $x > 4$

46. Which is the solution to $5 + 2x > 7$?

- (a) $x > 3$ (b) $x > 2$ (c) $x > 1$ (d) $x > 6$

Ans: (c) $x > 1$

47. Which is the solution to $x / 3 - 1 / 6 > 5 / 6$?

- (a) $x > 1$ (b) $x > 2$ (c) $x > 3$ (d) $x > 6$

Ans: (a) $x > 1$

48. Which inequality is true?

- (a) $4(6) > 26$ (b) $6+11 < 4$ (c) $10 - 6 < 5$ (d) $6+(-1) > 7$

Ans: (c) $10 - 6 < 5$

49. Solve $a + 3 \geq 2$

- (a) $a \leq -1$ (b) $a \geq -1$ (c) $a \geq 1$ (d) $a \geq 5$

Ans: (b) $a \geq -1$

50. Solve $a - 6 \geq -1$

- (a) $a \leq 5$ (b) $a \geq -5$ (c) $a \leq -5$ (d) $a \geq 5$

Ans: (d) $a \geq 5$

51. Which is the solution to $-3 \leq x / 3$?

- (a) $x \leq -9$ (b) $x \geq -9$ (c) $x \leq -1$ (d) $x \geq -1$

Ans: (b) $x \geq -9$

52. Solve $-8x > 48$

- (a) $x > -6$ (b) $x < -6$ (c) $x < 6$ (d) $x > 8$

Ans: (b) $x < -6$

53. Which is the solution to $a / 2 + 2 / 5 \leq 9 / 10$?

- (a) $a \leq 1$ (b) $a \leq 2$ (c) $a \leq 5$ (d) $a \leq 10$

Ans: (a) $a \leq 1$

54. Which inequality is true?

- (a) $5(-2) > 10$ (b) $11 + (-3) < 9$ (c) $3(13) < 39$ (d) $5 + 12 > 18$

Ans: (b) $11 + (-3) < 9$

55. Solve $-1 < a - 3$

- (a) $a > -4$ (b) $a > -2$ (c) $a > 1 / 3$ (d) $a > 2$

Ans: (d) $a > 2$

56. Solve $a - 6 \geq -6$

- (a) $a \leq -12$ (b) $a \geq -6$ (c) $a \geq 0$ (d) $a \geq 1$

Ans: (c) $a \geq 0$

57. Which is the solution to $-2 \leq (-2x / 3)$

- (a) $x \geq -3$ (b) $x \leq -3$ (c) $x \geq -9$ (d) $x \leq -9$

Ans: (a) $x \geq -3$

58. Solve $-10x < -100$

- (a) $x > 10$ (b) $x < 10$ (c) $x > -10$ (d) $-x > -10$

Ans: (a) $x > 10$

59. If $5x + 10 < 20$, what is the value of x ?

- (a) $x < 2$ (b) $x < 25$ (c) $x < 2/5$ (d) $x < 1$

Ans: (a) $x < 2$

60. Find the solution set of $x < 0$ from the set of integers $\{-7, -6, -5, -3, -1, 0, 1, 3, 4\}$.

- (a) $\{-5, -3\}$ (b) $\{-7, -6, -5, -3\}$ (c) $\{-7, -6, -5, -3, -1\}$ (d) $\{-7, -6, -5, -3, 0\}$

Ans: (c) $\{-7, -6, -5, -3, -1\}$

61. The equation for the statement "5 less than thrice a number x is 7" is:

- (a) $3x - 5 = 7$ (b) $5 - 3x = 7$ (c) $7 - 3x = 5$ (d) $3x + 5 = 7$

Ans: (a) $3x - 5 = 7$

62. Which of the following is an incorrect inequality?

- (a) $24 \div (-6) > -25 \div 5$ (b) $(-8) + 11 = 0 \div (100) + 11$
(c) $8 \div (-4) - 1 > 8 \div 4 - 1$ (d) $(-100) \div 25 \div (-1) = 100 \div (-25) \div (-1)$

Ans: (c) $8 \div (-4) - 1 > 8 \div 4 - 1$

63. A number, such that one-fifth of the number is 3 more than 7. What is the number?

- (a) 40 (b) 50 (c) 30 (d) 20

Ans: (b) 50

$$\frac{1}{5}x = 7 + 3 \Rightarrow \frac{1}{5}x = 10 \Rightarrow x = 50$$

64. Factors of the terms $-4pq^2$ in the expression $9p^2q^2 - 4pq^2$ are

- (a) $9p^2q^2, -4pq^2$ (b) 9, -4 (c) -4, p, q, q (d) -4

Ans: (c) -4, p, q, q

65. If the length of each side of the equilateral triangle is l , then the perimeter of the equilateral triangle is

- (a) $3l$ (b) $3 + l$ (c) $3 - l$ (d) $l / 3$

Ans: (a) $3l$

66. Which of the following is monomial

- (a) $2x + 3$ (b) $2x$ (c) $4x + 2y + 3$ (d) $4y + 5x + z - 1$

Ans: (b) $2x$

67. Which of the following is trinomial

- (a) $2a + 6b - 1$ (b) 1 (c) $5a - 7$ (d) $a + b + c - 3$

Ans: (d) $a + b + c - 3$

68. Terms with factors y in the expression $8 + xy + xyz$ are

- (a) xy, xyz (b) x, xz (c) $8, xy, xyz$ (d) y, xz

Ans: (a) xy, xyz

69. Identify the terms in the expression $x + y + 1$ which are not constant

- (a) $x, y, 1$ (b) x, y (c) $x, 1$ (d) $y, 1$

Ans: (b) x, y

70. The value of expression $4x - 3$ at $x=2$ is

- (a) -4 (b) 5 (c) 4 (d) 2

Ans: (b) 5

71. The value of expression $5n^2 + 5n - 2$ for $n = -2$ is

- (a) 13 (b) 3 (c) 8 (d) 12

Ans: (c) 8

72. The value of expression $2a^2 + 2b^2 - ab$ for $a = 2, b = 1$ is

- (a) 2 (b) 8 (c) 6 (d) 10

Ans: (b) 8

73. The value of $x + 7 + 4(x - 5)$ for $x=2$

- (a) -3 (b) 31 (c) 12 (d) 37

Ans: (a) -3

74. The value of expression $2a - 2b - 4 - 5 + a$ at $a=1, b = -2$

- (a) 10 (b) -2 (c) 12 (d) -4

Ans: (b) -2

75. What must be subtracted from $2a + b$ to get $2a - b$

- (a) $2b$ (b) $4a$ (c) 0 (d) $4a+4b$

Ans: (a) $2b$

76. What must be added to $3x + y$ to get $2x + 3y$

- (a) $5x + 4y$ (b) $-x + 2y$ (c) $x - 2y$ (d) $x+2y$

Ans: (b) $-x + 2y$

77. Subtract $a + 2b$ from sum of $a - b$ and $2a+b$

- (a) $2a - 2b$ (b) $4a + 2b$ (c) $2b$ (d) $-2a + 2b$

Ans: (a) $2a - 2b$

78. On simplifying $(a + b - 3) - (b - a + 3) + (a - b + 3)$ the result is

- (a) $a - b + 3$ (b) $a - b - 3$ (c) $3a - b - 3$ (d) $3a+b+3$

Ans: (c) $3a - b - 3$

79. What should be value of 'a' if $y^2 + y - a$ equals to 3 for $y=1$

- (a) -1 (b) -5 (c) 5 (d) 0

Ans: (a) -1

80. What is an expression for the statement: "p is multiplied by 16"

- (a) $16p$ (b) $p/16$ (c) $p+16$ (d) $p-16$

Ans: (a) $16p$

81. The expression for the statement: "y multiplied by 10 and then 7 added to product".

- (a) $10 + y + 7$ (b) $7y + 10$ (c) $10y + 7$ (d) $10y$

Ans: (c) $10y + 7$

82. What is the statement for the expression $2y - 9$
(a) $2y$ subtracted from (b) 9 subtracted from y and multiplied by 2
(c) 9 subtracted from 9 (d) Thrice of y minus 9

Ans: (b) 9 subtracted from y and multiplied by 2

83. Give expression for: "5 times of ' y ' to which 3 is added"
(a) $y + 15$ (b) $5y + 3$ (c) $5 / y + 3$ (d) $3y + 5$

Ans: (b) $5y + 3$

84. The equation for the statement: one fourth of a number minus 4 gives 4.

- (a) $4x - 4 = 4$ (b) $(4 / x) - 4 = 4$
(c) $(1 / 4) x - 4 = 4$ (d) $x - 4 = 1 / 4$

Ans: (c) $(1 / 4) x - 4 = 4$

85. What is the product of $3x$ and $(x^2 - 2x)$?

- (a) $3x^2 - 2x$ (b) $3x^3 - 2x^2$ (c) $3x^3 - 6x^2$ (d) $x^2 - 2x$

Ans: (c) $3x^3 - 6x^2$

86. Find the product: $2xy(x^2 - y^2)$

- (a) $2x^3y - 2xy^2$ (b) $2x^3y - 2xy^3$ (c) $x^2y - xy^2$ (d) $2x^2y - 2xy^2$

Ans: (b) $2x^3y - 2xy^3$

87. The product of $(2a - 6b)$ and $(a + 4b)$ is

- (a) $2a^2 + 8ab - 6ab - 24b^2$ (b) $2a^2 + 2ab - 24b^2$
(c) $2a^2 - 2ab - 24b^2$ (d) $2a^2 + 2ab + 24b^2$

Ans: (b) $2a^2 + 2ab - 24b^2$

88. What is the value of $5xy \times (-2x - 5y)$?

- (a) $-10x^2y - 25xy^2$ (b) $10x^2y + 25xy^2$ (c) $-10x^2y - 25y^2x$ (d) $5xy$

Ans: (a) $-10x^2y - 25xy^2$

89. Simplify: $(4abc + 3ab) \times (-4b^2c)$

- (a) $-16ab^2c^2 - 12ab^3c$ (b) $16ab^2c^2 + 12ab^3c$
(c) $-16ab^3c - 12ab^3c$ (d) $-16a^2b^2c^2 - 12a^2b^3c^2$

Ans: (c) $-16ab^3c - 12ab^3c$

90. Divide $-10x^3y^2$ by $2xy$.

- (a) $-5x^2y$ (b) $-5x^3y^2$ (c) $5x^2y$ (d) $-5xy$

Ans: (a) $-5x^2y$

91. Simplify: $6x^2y^3 / 3xy$

- (a) $2xy^2$ (b) $2x^2y^2$ (c) $2x^2y^3$ (d) $2xy$

Ans: (a) $2xy^2$

92. The result of dividing $(4p^2 - 2p)$ by $2p$ is:

- (a) $2p - 1$ (b) $2p^2 - 1$ (c) $4p^2 - 2$ (d) $2p$

Ans: (a) $2p - 1$

93. Find the value of $-12a^2b^3c$ by $4abc$.

- (a) $-3ab^2c$ (b) $-3ab^2$ (c) $3ab^2$ (d) $3ab^2c$

Ans: (b) $-3ab^2$

94. What is the value of $15x^3y^2 - 10x^2y^3 + 5x^2y^2$ by $5x^2y^2$?

- (a) $3x - 2y + 1$ (b) $3x - 2y$ (c) $15x - 10y + 5$ (d) $3x + 2y + 1$

Ans: (a) $3x - 2y + 1$

95. Which of the following statements is a valid inequality?

- (a) $5x + 2 = 17$ (b) $3x - 4 > 10$ (c) $6x + 9$ (d) $2y - 8$

Ans: (b) $3x - 4 > 10$

96. Find the value of the expression $x^3 - (1/x^3)$ if $x - 1/x = 10$ is

- (a) 1000 (b) 1030 (c) 970 (d) 100

Ans: (b) 1030

97. The difference between two numbers is 11. Six times the smaller number plus the larger number is 74. Find the numbers.

- (a) 10 and 21 (b) 9 and 20 (c) 12 and 23 (d) 15 and 26

Ans: (a) 10 and 21

98. If $x=3$, evaluate the expression $2x^2 + 5x - 1$.

- (a) 28 (b) 32 (c) 34 (d) 30

Ans: (c) 34

99. Which two-digit number is such that the sum of its digits is 9 and when reversed it increases by 27?

- (a) 36 (b) 37 (c) 35 (d) 34

Ans: (a) 36

100. What two digit number is 5 times the sum of its digits and is also equal to 5 more than twice the product of its digit?

- (a) 44 (b) 45 (c) 46 (d) 47

Ans: (b) 45